FAIRagro

Research Data

What is Research Data?

Research Data is generated during scientific work (e.g. through measurements, surveys, source work) and is the basis of scientific work and document its results. [forschungsdaten.info]

Nevertheless, there are different definitions depending on:

- Discipline
- Research Funders



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Source: ZB MED 2023, forschungsdaten.info, Illustration: “definition”” from Dewi Novita Sari licensed under CC BY 3.0

So many “types” of data…

Data status: describes the processing status of data

- *Examples:* raw data, processed data, aggregated data, …

Data origin: where does the data comes from?

- *Examples:* primary data, secondary data

Data category: research data type / data domain

- *Examples:* time series data, genetic data, phenotypic data, sensor data, satellite data, drone data, legacy data, …

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Research Data Management

Thought experiment

This is Lena.
She is an agricultural scientist



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Lena is calling **you**!
She wants some of your **data**



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Illustration: “telephone”” by popcornart and “biologist”” by Amethyst Studio are licensed under CC BY 3.0

Thought experiment

- ➔ Who can **find** the data Lena is looking for **in two minutes**?
- ➔ Will Lena be able to **open** the data?
- ➔ Will she be able to **understand** the data?
  - and you, do you still understand your own data?



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Illustration: "biologist"" by Amethyst Studio from noun project is licensed under CC BY 3.0

What does Lena need to be able to use the data?

Backup



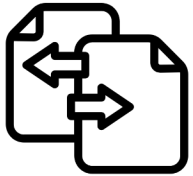
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Data-organisation



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Inter-operability



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Metadata



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Docu-mentation



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She needs good Research Data Management

Illustration: “File Edit”“, “File Transfer”“, “File Tag”“, “Folder Backup”“ and “Folder Structure”“ by Rafia Rauf from noun project are licensed under CC BY 3.0

What is Research Data Management (RDM)?

Series of measures that need to be taken during a research project in order to:

1. Obtain **high-quality data**
2. Make the **data available** and **usable** over the long-term
3. Make research findings **reproducible** beyond the research project



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Why RDM is important - for science

- ➔ Essential part of good scientific practice
  - 11 of the DFG's 19 guidelines (also) describe the handling of research data
- ➔ Reusable research data saves money, time and resources
- ➔ Ensures transparency, verifiability and reproducibility



Source: forschungsdaten.info; Illustration: Deutsche Forschungsgemeinschaft (2022). Guidelines for Safeguarding Good Research Practice. Code of Conduct

Why RDM is important - for you

- Eligibility for financing
- Saves time, nerves and resources
- Facilitates the induction of new employees
- Enhances teamwork and collaborations
- Helps meet formal and legal requirements
- Helps to keep track of the project
- Guaranteeing transparency, verifiability and reproducibility
- Citations, citations, citations, ...



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Source: Modified from Assmann et al. 2022, Bobrov et al. 2021, Bres et al. 2022, Jacob et al. 2022,
Lindstädt et al. 2019 und Voigt et al. 2022 von ZB MED 2024, Illustration: "botanist"" by Amethyst Studio is licensed under CC BY 3.0

FAIRagro

Research Data Life Cycle

What is the Research Data Life Cycle?

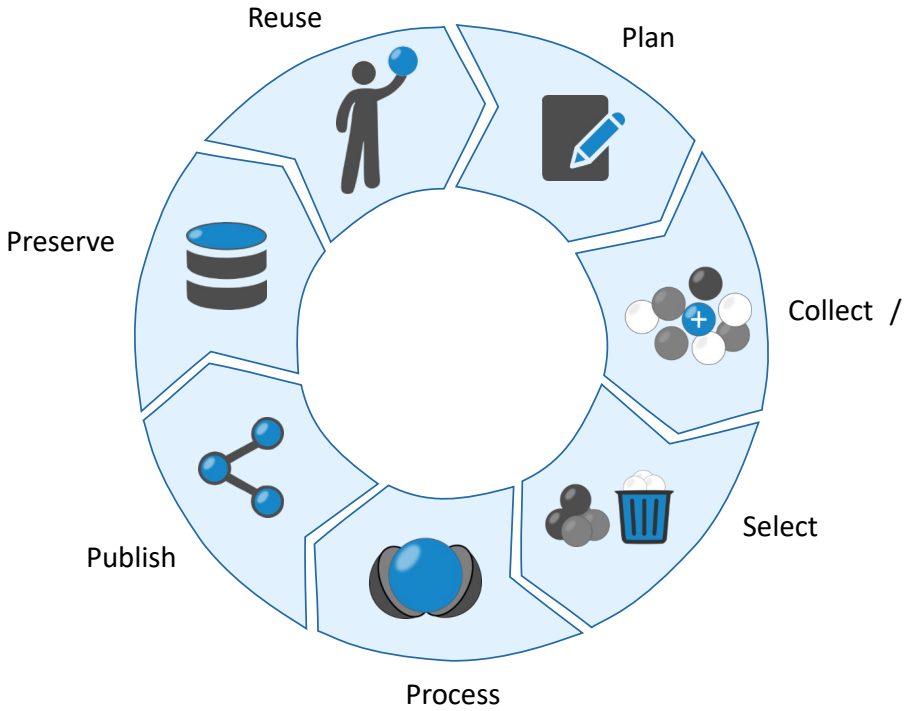
Model that illustrates the steps of research data management and **describes** how data should ideally flow through a research project to **ensure** successful data **curation** and **preservation**.



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Sources: Nanyang Technological University and Princeton Research Data Service modified by ZB MED 2023; Illustration “definition” by Dewi Novita Sari is licensed under CC BY 3.0

Research Data Life Cycle



Research Data Life Cycle

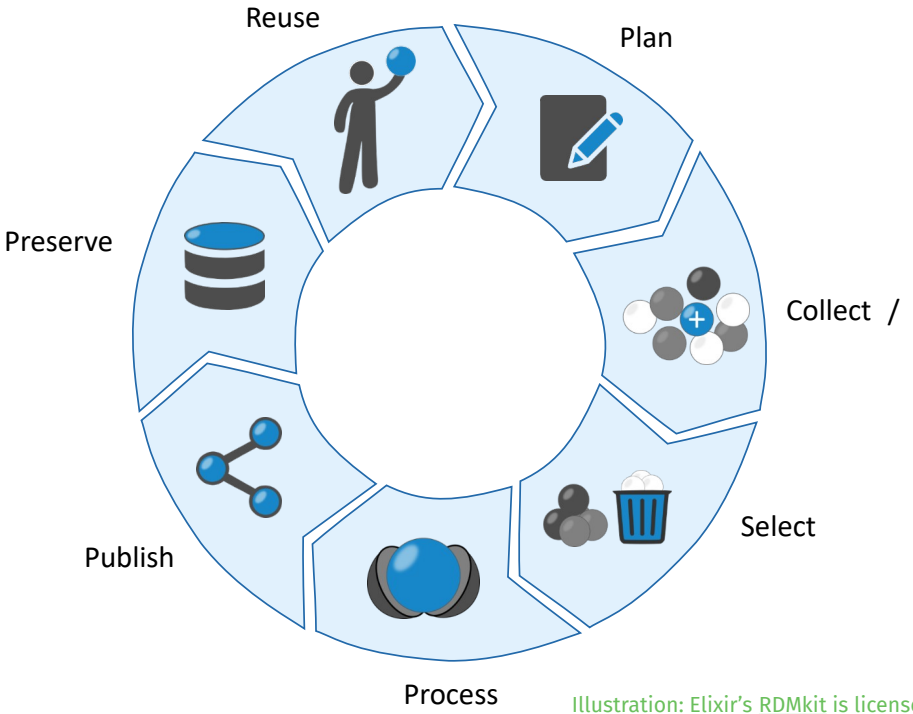




Illustration: Elixir's RDMkit is licensed under CC BY 4.0

Icebreaker

Please note down

- ➔ Your Name, your institute
- ➔ The Datatype you are working with

  When it's your turn
- ➔ Pin it to the stage of the Lifecycle, you are in at the moment



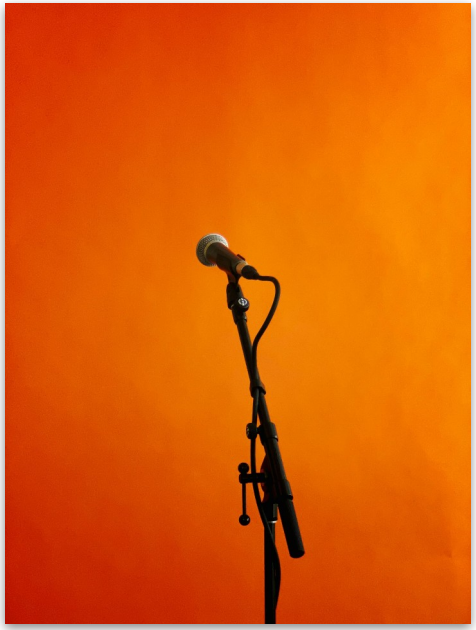


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